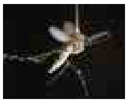

Mosquitoes remember swatters

Research has shown that mosquitoes can remember who swatted them by association with the distinct odour. http://dgit.in/mozmemory

Caltech building robots with Disney

Caltech and Disney have announced a collaborative effort to develop robots, an early prototype shows a bouncing ball bot. http://dgit.in/discalbot

Climate Engineering

It could be our last and most effective hope to undo the effects of global warming

Arnab Mukherjee | arnab@digit.in

While some people still debate the veracity of climate change data, most scientists (read, smart people), have not only accepted it, but have a very bleak outlook. While on the one hand the US pulls out of the Paris accord, on the other, you have studies telling us that we have failed miserably, and that mitigation and mere reductionary tactics aren't going to cut it anymore.

There's a need for something more drastic, and climate engineering might just be that solution! It involves us directly and intentionally intervening with the climate system using a host of cutting edge technologies. Yes, we know it sounds like the beginning of a sci-fi novel, but it's not as far fetched as it sounds.

BROAD APPROACHES

Before we begin, let's make one thing clear – climate engineering should never be viewed as a substitute for climate change mitigation. It is an add-on to help us when we fail to meet targets, and when disasters such as the US going rogue happen. Climate engineering solutions can be broadly divided into two kinds of approaches: solar radiation management and carbon dioxide removal. As in, removing carbon from our atmosphere and keeping some away from the earth.

CO_2 REMOVAL

At this point, through climate engineering, experiments have aimed to remove CO_2 from the atmosphere and park it in the oceans, terrestrial biosphere, geological reservoirs, or commercial materials. In this area, there are a couple of approaches that we've tried, or are trying now.

Ocean fertilisation using iron

This approach involves cultivating phytoplankton on the surface of the ocean by dropping iron into the water bodies. These phytoplanktons, spread across the ocean surface, will absorb atmospheric CO_2 and release oxygen. The problem with this process is that once the phytoplankton die en masse, digesting bacteria that are attracted to their bodies can rid a large portion of the ocean of its oxygen, rendering it unlivable for most aquatic life. Such a phenomenon has already happened in the Gulf of Mexico due to nitrates from fertilizers running down the Mississippi River inducing dead zones that cover as much as 22,500 square kilometers!

Direct air capture (DAC)

A more effective approach, with fewer side effects, involves sucking the CO_2 directly out of the atmosphere and putting it to other uses. While the obvious place to start seems to be at major generation sites like factories with post-combustion gathering, this wouldn't affect the CO_2 already in the atmosphere much – which is our primary concern at this point.

Direct removal of CO_2 has to be combined with the long term storage to be effective in bringing down the level of carbon dioxide in the atmosphere. There are a couple of techniques to do that.

For instance, using a chemical process known as carbonation, CO_2 can be bound to substrates. There are other chemical reactions that can be employed for this purpose too. And all of these have potential real life applications. Projects include an artificial tree concept that can convert atmospheric CO_2 to bicarbonates as well as an operation commercial plant in Switzerland by Climeworks AG that filters atmospheric CO_2 and supplies it to buyers.

A "red tide" off the coast of La Jolla, San Diego, California.

SOLAR RADIATION MANAGEMENT

Our only true source of energy in our solar system is our sun, and thanks to the proliferation of greenhouse gases and the subsequent depletion of the ozone layer, we've been getting a little too much of that energy. This is where solar radiation management (SRM) comes in as an approach to bring down the temperature of the planet. Overall, its objective is to reflect some of the